COMPUTER NETWORKING AND CYBER SECURITY

Training Modules Covered (1 Year)

- Networking Basics
- CCNA (Cisco Certified Network Associate)
- Microsoft Server
- Linux (Basic)
- Cloud Computing (AWS)
- Basic Python
- Ethical Hacking & Cyber Security

Benefits

- Training conducted by Certified & Experienced Trainers/Consultants
- Study Material provided by Trainers in soft copy
- Course Certificate Provided on completion of Training
- Membership for 1.5 Year for clearing doubt
- EMIs Option
- Weekdays (4 days) & Weekends (2 days-Sat & Sun) Class schedule
- Live Online & Offline Training Mode
- Placement Assistance with regular hiring by IT companies from DUCAT campus drive

COURSE CURRICULUM

NETWORKING BASICS

- Network Theory
- Network Communicaons Methods
- Network Media and Hardware
- Bounded Network Media
- Unbounded Network Media
- Noise Control
- Network Connecvity Devices
- Network Implementaons
- Ethernet Networks
- Wireless Networks
- The OSI Model
- The TCP/IP Model
- TCP/IP Addressing and Data Delivery
- The TCP/IP Protocol Suite
- IP Addressing
- Default IP Addressing Schemes
- Create Custom IP Addressing Schemes
- Implement IPv6 Addresses
- Delivery Techniques
- TCP/IP Services
- Assign IP Addresses

- Domain Naming Services
- TCP/IP Commands
- Common TCP/IP Protocols
- TCP/IP Interoperability Services
- LAN Infrastructure
- Switching
- Enable Stac Roung
- Implement Dynamic IP Roung
- Plan a SOHO Network

CCNA

Network Fundamentals

- Explain the role and funcon of network components
- Routers
- L2 and L3 switches
- Next-generaon firewalls and IPS
- Access points
- Controllers (Cisco DNA Center and WLC)
- Endpoints
- Servers

Describe characteristics of network topology architectures

- 2 tier
- 3 er
- Spine-leaf
- WAN
- Small office/home office (SOHO)
- On-premises and cloud

Compare physical interface and cabling types

- Single-mode fiber, mulmode fiber, copper
- Connecons (Ethernet shared media and point-to-point)
- Concepts of PoE

Identify interface and cable issues (collisions, errors, mismatch duplex, and/or speed)

Compare TCP to UDP

Configure and verify IPv4 addressing and subnetting

Describe the need for private IPv4 addressing

Configure and verify IPv6 addressing and prefix

Compare IPv6 address types 1.9.a Global unicast

- Unique local
- Link local
- Anycast
- Mulcast
- Modified EUI 64

Verify IP parameters for Client OS (Windows, Mac OS, Linux)

Describe wireless principles

- Nonoverlapping Wi-Fi channels
- SSID
- RF
- Encrypon

Explain virtualization fundamentals (virtual machines)

Describe switching concepts

- MAC learning and aging
- Frame switching

- Frame flooding
- MAC address table

2.0 Network Access

- Configure and verify VLANs (normal range) spanning mulple switches
- Access ports (data and voice)
- Default VLAN
- Connecvity

Configure and verify interswitch connectivity

- Trunk ports
- 802.1Q
- Nave VLAN

Configure and verify Layer 2 discovery protocols (Cisco Discovery Protocol and LLDP)

Configure and verify (Layer 2/Layer 3) EtherChannel (LACP)

Describe the need for and basic operations of Rapid PVST+ Spanning Tree Protocol and identify basic operations

- Root port, root bridge (primary/secondary), and other port names
- Port states (forwarding/blocking)
- PortFast benefits

Compare Cisco Wireless Architectures and AP modes

Describe physical infrastructure connections of WLAN components (AP, WLC, access/trunk ports, and LAG)

Compare Cisco Wireless Architectures and AP modes

Describe physical infrastructure connections of WLAN components (AP, WLC, access/trunk ports, and LAG)

Describe AP and WLC management access connections (Telnet, SSH, HTTP, HTTPS, console, and TACACS+/RADIUS)

Configure the components of a wireless LAN access for client connectivity using GUI only such as WLAN creation, security settings, QoS profiles, and advanced WLAN settings

3.0 IP Connectivity

Interpret the components of routing table

- Roung protocol code
- Prefix
- Network mask
- Next hop
- Administrave distance
- Metric
- Gateway of last resort

Determine how a router makes a forwarding decision by default

- Longest match
- Administrave distance
- Roung protocol metric

Configure and verify IPv4 and IPv6 static routing

- Default route
- Network route
- Host route
- Floang stac

Configure and verify single area OSPFv2 3.4.a Neighbor adjacencies

- Point-to-point
- Broadcast (DR/BDR selecon)
- Router ID

Describe the purpose of first hop redundancy protocol

4.0 IP Services

- Configure and verify inside source NAT using stac and pools
- Configure and verify NTP operang in a client and server mode
- Explain the role of DHCP and DNS within the network
- Explain the funcon of SNMP in network operaons
- Describe the use of syslog features including facilies and levels
- Configure and verify DHCP client and relay
- Explain the forwarding per-hop behavior (PHB) for QoS such as classificaon, marking, queuing, congeson, policing, shaping
- Configure network devices for remote access using SSH
- Describe the capabilies and funcon of TFTP/FTP in the network

5.0 Security Fundamentals

- Define key security concepts (threats, vulnerabilies, exploits, and migaon techniques)
- Describe security program elements (user awareness, training, and physical access control)
- Configure device access control using local passwords
- Describe security password policies elements, such as management, complexity, and password alternaves (mulfactor authencaon, cerficates, and biometrics)
- Describe remote access and site-to-site VPNs
- Configure and verify access control lists
- Configure Layer 2 security features (DHCP snooping, dynamic ARP inspecon, and port security)
- Differenate authencaon, authorizaon, and accounng concepts
- Describe wireless security protocols (WPA, WPA2, and WPA3)
- Configure WLAN using WPA2 PSK using the GUI

6.0 Automation and Programmability

- Explain how automaon impacts network management
- Compare tradional networks with controller-based networking
- Describe controller-based and soware defined architectures (overlay, underlay, and fabric)
- Separaon of control plane and data plane
- North-bound and south-bound APIs

Compare traditional campus device management with Cisco DNA Center enabled device management

Describe characteristics of REST-based APIs (CRUD, HTTP verbs, and data encoding)

Recognize the capabilities of configuration management mechanisms Puppet, Chef, and Ansible

Interpret JSON encoded data

MICROSOFT SERVER-2016/2019

Implement Domain Name System (DNS)

Install and configure DNS servers

- This objective may include but is not limited to: Determine DNS installation requirements; install DNS; configure forwarders; configure Root Hints; configure delegation; implement DNS policies; Configure DNS Server settings using Windows PowerShell; configure Domain Name System Security Extensions (DNSSEC); configure DNS Socket Pool; configure cache locking; enable Response Rate Limiting; configure DNS-based Authentication of Named Entities (DANE); configure DNS logging; configure delegated administration; configure recursion settings; implement DNS performance tuning; configure global settings

Create and configure DNS zones and records

- This objective may include but is not limited to: Create primary zones; configure Active Directory primary zones; create and configure secondary zones; create and configure stub zones; configure a GlobalNames zone; analyze zone-level statistics; create and configure DNS Resource Records (RR), including A, AAAA, PTR, SOA, NS, SRV, CNAME, and MX records; configure zone scavenging; configure record options, including Time To Live (TTL) and weight; configure round robin; configure secure dynamic updates; configure unknown record support; use DNS audit events and analytical (query) events for auditing and troubleshooting; configure Zone Scopes; configure records in Zone Scopes; configure policies for zones

Create and configure DNS zones and records

- This objective may include but is not limited to: Create primary zones; configure Active Directory primary zones; create and configure secondary zones; create and configure stub zones; configure a GlobalNames zone; analyze zone-level statistics; create and configure DNS Resource Records (RR), including A, AAAA, PTR, SOA, NS, SRV, CNAME, and MX records; configure zone scavenging; configure record options, including Time To Live (TTL) and weight; configure round robin; configure secure dynamic updates; configure unknown record support; use DNS audit events and analytical (query) events for auditing and troubleshooting; configure Zone Scopes; configure records in Zone Scopes; configure policies for zones

Implement DHCP and IPAM

Install and configure DHCP

- This objective may include but is not limited to: Install and configure DHCP servers; authorize a DHCP server; create and configure scopes; create and configure superscopes and multicast scopes; configure a DHCP reservation; configure DHCP options; configure DNS options from within DHCP; configure policies; configure client and server for PXE boot; configure DHCP Relay Agent; implement IPv6 addressing using DHCPv6; perform export and import of a DHCP server; perform DHCP server migration.

Manage and maintain DHCP

- This objective may include but is not limited to: Configure a lease period; back up and restore the DHCP database; configure high availability using DHCP failover; configure DHCP name protection; troubleshoot DHCP.

Implement and Maintain IP Address Management (IPAM)

- This objective may include but is not limited to: Provision IPAM manually or by using Group Policy; configure server discovery; create and manage IP blocks and ranges; monitor utilization of IP address space; migrate existing workloads to IPAM; configure IPAM database storage using SQL Server; determine scenarios for using IPAM with System Center Virtual Machine Manager for physical and virtual IP address space management, Manage DHCP server properties using IPAM; configure DHCP scopes and options; configure DHCP policies and failover; manage DNS server properties using IPAM; manage DNS zones and records; manage DNS and DHCP servers in multiple Active Directory forests; delegate administration for DNS and DHCP using role-based access control (RBAC); Audit the changes performed on the DNS and DHCP servers; audit the IPAM address usage trail; audit DHCP lease events and user logon events.

Implement Network Connectivity and Remote Access Solutions

Implement network connectivity solutions

- This objective may include but is not limited to: Implement Network Address Translation (NAT); configure routing.

Implement virtual private network (VPN) and DirectAccess solutions

- This objective may include but is not limited to: Implement remote access and site-to-site (S2S) VPN solutions using remote access gateway; configure different VPN protocol options; configure authentication options; configure VPN reconnect; create and configure connection profiles; determine when to use remote access VPN and site-to-site VPN and configure appropriate protocols; install and configure DirectAccess; implement server requirements; implement client configuration; troubleshoot DirectAccess

Implement Network Policy Server (NPS)

- This objective may include but is not limited to: Configure a RADIUS server including RADIUS proxy; configure RADIUS clients; configure NPS templates; configure RADIUS accounting; configure certificates; configure Connection Request Policies; configure network policies for VPN and wireless and wired clients; import and export NPS policies

Implement Core and Distributed Network Solutions

Implement IPv4 and IPv6 addressing

- This objective may include but is not limited to: Configure IPv4 addresses and options; determine and configure appropriate IPv6 addresses; configure IPv4 or IPv6 subnetting; implement IPv6 stateless addressing; configure interoperability between IPv4 and IPv6 by using ISATAP, 6to4, and Teredo scenarios; configure Border Gateway Protocol (BGP); configure IPv4 and IPv6 routing.

Implement Distributed File System (DFS) and Branch Office solutions

- This objective may include but is not limited to: Install and configure DFS namespaces; configure DFS replication targets; configure replication scheduling; configure Remote Differential Compression (RDC) settings; configure staging; configure fault tolerance; clone a Distributed File System Replication (DFSR) database; recover DFSR databases; optimize DFS Replication; install and configure BranchCache; implement distributed and hosted cache modes; implement BranchCache for web, file, and application servers;; troubleshoot BranchCache.

Implement an Advanced Network Infrastructure

Implement high performance network solutions

- This objective may include but is not limited to: Implement NIC Teaming or the Switch Embedded Teaming (SET) solution and identify when to use each; enable and configure Receive Side Scaling (RSS); enable and configure network Quality of Service (QoS) with Data Center Bridging (DCB); enable and configure SMB Direct on Remote Direct Memory Access (RDMA) enabled network adapters; configure SMB Multichannel; enable and configure virtual Receive Side Scaling (vRSS) on a Virtual Machine Queue (VMQ) capable network adapter; enable and configure Virtual Machine Multi-Queue (VMMQ); enable and configure Single-Root I/O Virtualization (SR-IOV) on a supported network adapter.

Determine scenarios and requirements for implementing Software Defined Networking (SDN)

- This objective may include but is not limited to: Determine deployment scenarios and network requirements for deploying SDN; determine requirements and scenarios for implementing Hyper-V Network Virtualization (HNV) using Network Virtualization Generic Route Encapsulation (NVGRE) encapsulation or Virtual Extensible LAN (VXLAN) encapsulation; determine scenarios for implementation of Software Load Balancer (SLB) for North-South and East-West load balancing; determine implementation scenarios for various types of Windows Server Gateways, including L3, GRE, and S2S, and their use; determine requirements and scenarios for Datacenter firewall policies and network security groupson a supported network adapter.

LINUX (BASIC)

Linux System Administraon - I

- Geng started with Red Hat Enterprise Linux
- Access the command line
- Manage files from the command line
- Get help in Red Hat Enterprise Linux
- Create, view & edit text files
- Manage local users & groups
- Control access to files
- Monitor & manage Linux processes
- Control services & daemons
- Configure & secure SSH
- Analyze & store logs
- Manage networking
- Archive & transfer files
- Install & update soware
- Access Linux file systems

- Analyze servers & get support
- Comprehensive review

Linux System Administraon - II

- Improving command line producvity using shell scripts
- Schedule future tasks
- Tune system performance
- Manage SELinux security
- Maintain basic storage
- Manage storage stack
- Access network-aached storage
- Control the boot process
- Manage network security
- Install Red Hat Enterprise Linux
- Run Containers
- Comprehensive Review

AWS

1. Introduction to Cloud Computing:

In this module, you will learn what Cloud Computing is and what are the different models of Cloud Computing along with the key differentiators of different models. We will also introduce you to virtual world of AWS along with AWS key vocabulary, services and concepts.

- Introduction to Cloud Computing
- AWS Architecture
- AWS Management Console
- Setting up of the AWS Account

2. Amazon EC2 and Amazon EBS:

In this module, you will learn about the introduction to compute offering from AWS called EC2. We will cover different instance types and Amazon AMIs. A demo on launching an AWS EC2 instance, connect with an instance and hosting a website on AWS EC2 instance. We will also cover EBS storage Architecture (AWS persistent storage) and the concepts of AMI and snapshots.

- Amazon EC2
- Amazon EBS
- Demo of AMI Creation
- Backup, Restore

3. Amazon Storage Services S3 (Simple Storage Services):

In this module, you will learn how AWS provides various kinds of scalable storage services. In this module, we will cover different storage services like S3, Glacier, Versioning, and learn how to host a static website on AWS.

4. Cloud Watch & SNS:

In this modules, you will learn how to monitoring AWS resources and setting up alerts and notifications for AWS resources and AWS usage billing with AWS CloudWatch and SNS.

- Amazon Cloud Watch
- SNS - Simple Notification Services

5. Scaling and Load Distribution in AWS:

In this module, you will learn about 'Scaling' and 'Load distribution techniques' in AWS. This module also includes a demo of Load distribution & Scaling your resources horizontally based on time or activity.

- Amazon Auto Scaling & ELB

6. AWS VPC & Route 53:

In this module, you will learn introduction to Amazon Virtual Private Cloud. We will cover how you can make public and private subnet with AWS VPC. A demo on creating VPC. We will also cover overview of AWS Route 53.

- Amazon VPC with subnets
- Gateways
- Route Tables
- Amazon Route 53 overview

7. Identity and Access Management Techniques (IAM):

In this module, you will learn how to achieve distribution of access control with AWS using IAM.

- Amazon IAM - add users to groups, manage passwords, log in with IAM-created users.

8. Amazon Relational Database Service (RDS):

In this module, you will learn how to manage relational database service of AWS called RDS.

- Amazon RDS

9. Multiple AWS Services and Managing the Resources' Lifecycle:

In this module, you will get an overview of multiple AWS services. We will talk about how do you manage life cycle of AWS resources and follow the DevOps model in AWS. We will also talk about notification and email service of AWS along with Content Distribution Service in this module.

- AWS CloudFront, Cloud Trail, SQS
- Overview of AWS Products such as Elastic Beanstalk, AWS DynamoDB Cloud Formation, SES.

10. AWS Architecture and Design:

In this module, you will cover various architecture and design aspects of AWS. We will also cover the cost planning and optimization techniques along with AWS security best practices, High Availability (HA) and Disaster Recovery (DR) in AWS

- AWS Backup and DR Setup
- AWS High Availability Design
- AWS Best Practices (Cost +Security)
- AWS Calculator & Consolidated Billing

11. Migrating to Cloud & AWS:

In this module, you will learn how to migrate to cloud.

- Amazon Web Services Certifications Training.

BASIC PYTHON

Introducon To Python

- Why Python
- Applicaonareas of python
- Pythonimplementaons

1. Cpython
2. Jython
3. Ironpython
4. Pypy

- Pythonversions
- Installingpython
- Pythoninterpreter architecture

5. Python byte code compiler
6. Python virtual machine(pvm)

Wring and Execung First Python Program

- Using interacve mode
- Using script mode

1. General text editor and command window
2. Idle editor and idle shell

- Understandingprint() funcon
- How to compile python program explicitly

Python Language Fundamentals

- Characterset
- Keywords
- Comments
- Variables
- Literals
- Operators
- Reading input from console
- Parsing string to int, float

Python Condional Statements

- If statement
- If else statement
- If elif statement
- If elif else statement
- Nested if statement

Looping Statements

- While loop
- For loop
- Nested loops
- Pass, break and connue keywords

Standard Data Types

- Int, float, complex, bool, nonetype
- Str, list,tuple,range
- Dict, set, frozenset

String Handling

- Whatis string
- String representaons
- Unicode string
- String funcons, methods
- String indexing and slicing
- String formang

Python List

- Creang and accessing lists
- Indexing and slicing lists
- List methods
- Nested lists
- List comprehension

Python Tuple

- Creang tuple
- Accessing tuple
- Immutabilityof tuple

Python Set

- How to create a set
- Iteraon over sets
- Pythonset methods
- Python frozenset

Python Diconary

- Creang a diconary
- Diconarymethods
- Accessing values from diconary
- Updang diconary

- Iterang diconary
- Diconarycomprehension

Python Funcons

- Defining a funcon
- Calling a funcon
- Typesof funcons
- Funcon arguments

1. Posional arguments, keyword arguments
2. Default arguments, non-default arguments
3. Arbitrary arguments, keyword arbitrary arguments

- Funcon return statement
- Nested funcon
- Funcon as argument
- Funcon as return statement
- Decoratorfuncon
- Closure
- Map(),filter(), reduce(), any() funcons
- Anonymous or lambda funcon

Modules & Packages

- Why modules
- Script v/s module
- Imporngmodule
- Standardv/s third party modules
- Why packages
- Understandingpip ulity

File I/O

- Introduconto file handling
- File modes
- Funcons and methods related to file handling
- Understandingwith block

Regular Expressions(Regex)

- Need of regular expressions
- Re module
- Funcons /methods related to regex
- Meta characters & specialsequences

Object Oriented Programming

- Procedural v/s Object Oriented Programming
- OOP Principles
- Defining a Class & Object Creaon
- Inheritance
- Encapsulaon
- Polymorphism
- Abstracon
- Garbage Collecon
- Iterator & Generator

Excepon Handling

- Difference Between Syntax Errors and Excepons
- Keywords used in Excepon Handling

1. try , except , finally , raise , assert

- Types of Except Blocks
- User-defined Excepons

GUI Programming

- Introducon to Tkinter Programming
- Tkinter Widgets
- Layout Managers
- Event handling
- Displaying image

Mul Threading Programming

- Mul-processing v/s Mul-threading
- Need of threads
- Creang child threads
- Funcons /methodsrelated to threads
- Thread synchronizaon and locking

ETHICAL HACKING

Introduction to Ethical Hacking

- Information Security Overview
- Information Security Threats and Attack Vectors
- Top Information Security Attack Vectors
- Motives, Goals, and Objectives of Information Security Attacks
- Information Security Threats
- Information Warfare
- Hacking Concepts
- Hacking vs. Ethical Hacking
- Effects of Hacking on Business
- Who Is a Hacker?
- Hacker Classes
- Hacktivism
- Hacking Phases
- Types of Attacks
- Types of Attacks on a System
- Operating System Attacks
- Misconfiguration Attacks
- Application-Level Attacks
- Skills of an Ethical Hacker
- Defense in Depth
- Incident Management Process
- Information Security Policies
- Classification of Security Policies
- Structure and Contents of Security Policies

Footprinting and Reconnaissance

- Foot printing Concepts
- Foot printing Terminology
- What is Foot printing?
- Why Foot printing?
- Objectives of Foot printing
- Foot printing Threats
- Foot printing through Search Engines
- Finding Company’s External and Internal URLs
- Mirroring Entire Website
- Website Mirroring Tools
- Extract Website Information from

- Monitoring Web Updates Using Website Watcher
- Finding Resources Using Google Advance Operator
- Google Hacking Tool: Google
- Hacking Database (GHDB)
- Google Hacking Tools
- WHOIS Footprinting
- WHOIS Lookup
- DNS Footprinting
- Extracting DNS Information
- DNS Interrogation Tools
- Network Footprinting
- Locate the Network Range
- Determine the Operating System
- Footprinting through Social Engineering

Scanning Networks

- Check for Live Systems
- Checking for Live Systems - ICMP Scanning
- Ping Sweep
- Check for Open Ports
- Scanning Tool: Nmap
- Hping2 / Hping3
- Scanning Techniques
- Scanning Tool: NetScan Tools Pro
- Scanning Tools
- Do Not Scan These IP Addresses
- (Unless you want to get into trouble)
- Port Scanning Countermeasures
- Banner Grabbing Countermeasures
- Disabling or Changing Banner
- Hiding File Extensions from Web Pages
- Scan for Vulnerability
- Proxy Servers
- Why Attackers Use Proxy Servers?
- Use of Proxies for Attack

Enumeration

- What is Enumeration?
- Techniques for Enumeration
- Services and Ports to Enumerate
- NetBIOS Enumeration
- NetBIOS Enumeration
- NetBIOS Enumeration Tool:SuperScan
- NetBIOS Enumeration Tool: Hyena
- NetBIOS Enumeration Tool:Winfingerprint
- NetBIOS Enumeration Tool:NetBIOS Enumerator
- Enumerating User Accounts

System Hacking

- Information at Hand Before System Hacking Stage
- System Hacking: Goals
- CEH Hacking Methodology (CHM)
- CEH System Hacking Steps
- CrackingPasswords

- Password Cracking
- Password Complexity
- Password Cracking Techniques
- Types of Password Attacks
- Distributed Network Attack
- Default Passwords
- Manual Password Cracking (Guessing)
- Stealing Passwords Using Keyloggers
- Spyware
- How to Defend Against Keyloggers
- Anti-Spywares
- What Is Steganography?
- Least Significant Bit Insertion

Trojans and Backdoors

- Trojan Concepts
- What is a Trojan?
- Trojan Infection
- Types of Trojans
- Command Shell Trojans
- Command Shell Trojan: Netcat
- GUI Trojan: MoSucker
- GUI Trojan: Jumper and Biodox
- Document Trojans
- E-mail Trojans
- E-mail Trojans: RemoteByMail
- Trojan Detection
- How to Detect Trojans
- Scanning for Suspicious Ports
- Trojan Horse Construction Kit
- Anti-Trojan Software

Viruses and Worms

- Virus and Worms Concepts
- Introduction to Viruses
- Virus and Worm Statistics
- Types of Viruses
- System or Boot Sector Viruses
- File and Multipartite Viruses
- Macro Viruses
- Cluster Viruses
- Stealth/Tunneling Viruses
- Encryption Viruses
- Polymorphic Code
- Computer Worms
- Malware Analysis
- Online Malware Testing: VirusTotal
- Online Malware Analysis Services
- Anti-virus Tools

Sniffers

- Sniffing Concepts
- Wiretapping
- Lawful Interception

- Packet Sniffing
- Sniffing Threats
- SPAN Port
- MAC Attacks
- MAC Flooding
- MAC Address/CAM Table
- How CAM Works
- DHCP Attacks
- How DHCP Works
- DHCP Request/Reply Messages
- IPv4 DHCP Packet Format
- ARP Poisoning
- What Is Address Resolution Protocol(ARP)?
- ARP Spoofing Techniques
- ARP Spoofing Attack
- Spoofing Attack
- Spoofing Attack Threats
- DNS Poisoning
- DNS Poisoning Techniques

Social Engineering

- Social Engineering Concepts
- What is Social Engineering?
- Behaviors Vulnerable to Attacks
- Social Engineering Techniques
- Types of Social Engineering
- Human-based Social Engineering
- Technical Support Example
- Authority Support Example
- Social Networking Sites
- Social Engineering Through
- Impersonation on Social
- Networking Sites
- How to Detect Phishing Emails
- Anti-Phishing Toolbar: Netcraft
- Anti-Phishing Toolbar: PhishTank
- Identity Theft Countermeasures

Denial of Service

- DoS/DDoS Concepts
- What is a Denial of Service Attack?
- What Are Distributed Denial ofService Attacks?
- Symptoms of a DoS Attack
- DoS Attack Techniques
- Bandwidth Attacks
- Service Request Floods
- SYN Attack
- SYN Flooding
- ICMP Flood Attack
- Peer-to-Peer Attacks
- Permanent Denial-of-Service Attack
- Application Level Flood Attacks
- Botnet

- Botnet Propagation Technique
- DDoS Attack
- DDoS Attack Tool:LOIC
- DoS Attack Tools

Session Hijacking

- Session Hijacking Concepts
- What is Session Hijacking?
- Dangers Posed by Hijacking
- Why Session Hijacking is Successful?
- Key Session Hijacking Techniques
- Brute Forcing Attack
- Network-level Session Hijacking
- The 3-Way Handshake
- Sequence Numbers
- Session Hijacking Tools
- Session Hijacking Tool: Zaproxy
- Session Hijacking Tool: Burp Suite
- Session Hijacking Tool: JHijack
- Session Hijacking Tools

Hacking Webservers

- Webserver Concepts
- Webserver Market Shares
- Open Source WebserverArchitecture
- Attack Methodology
- Webserver Attack Methodology
- Webserver Attack Methodology:Information Gathering
- Webserver Attack Methodology:Webserver Footprinting
- Counter-measures
- Countermeasures: Patches and Updates
- Countermeasures: Protocols
- Countermeasures: Accounts
- Countermeasures: Files and Directories
- How to Defend Against Web Server Attacks
- How to Defend against HTTP
- Response Splitting and Web Cache
- Poisoning
- Web Server Penetration Testing

Hacking Web Applications

- Web App Concepts
- Web Application Security Statistics
- Introduction to Web Applications
- SQL Injection Attacks
- Command Injection Attacks
- Web App Hacking Methodology
- Footprint Web Infrastructure
- Footprint Web Infrastructure:ServerDiscovery
- Hacking Web Servers
- Web Server Hacking Tool:WebInspect
- Web Services Probing Attacks
- Web Service Attacks: SOAP Injection
- Web Service Attacks: XML Injection

- Web Services Parsing Attacks
- Web Service Attack Tool: soapUI

SQL Injection

- SQLInjection Concepts
- SQL Injection
- Scenario
- SQL Injection Threats
- What is SQL Injection?
- SQL Injection Attacks
- SQL Injection Detection
- Types of SQL Injection
- Simple SQL Injection Attack
- Union SQL Injection Example
- SQL Injection Error Based
- Blind SQL Injection
- What is Blind SQL Injection?
- SQL Injection Methodology
- Advanced SQL Injection
- Information Gathering
- Extracting Information through ErrorMessages
- Interacting with the FileSystem
- SQL Injection Tools
- SQL Injection Tools: BSQLHacker
- SQL Injection Tools: Marathon Tool
- SQL Injection Tools: SQL PowerInjector
- SQL Injection Tools: Havij
- SQL Injection Tools

Hacking Wireless Networks

- Wireless Concepts
- Wireless Networks
- Wi-Fi Networks at Home and Public Places
- Types of Wireless Networks
- Wireless Encryption
- Wireless Threats
- Wireless Threats: Access Control Attacks
- Wireless Threats: Integrity Attacks
- Footprint the Wireless Network
- Attackers Scanning for Wi-Fi Networks
- Bluetooth Hacking
- Bluetooth Threats

Evading IDS, Firewalls, and Honeypots

- DS, Firewall and Honeypot Concepts
- How IDS Works?
- Ways to Detect an Intrusion
- Denial-of-Service Attack (DoS)
- ASCII Shellcode
- Other Types of Evasion
- Evading Firewalls
- IP Address Spoofing
- Source Routing

- Website Surfing Sites
- Detecting Honeypots

Buffer Overflow

- Buffer Overflow Concepts
- Buffer Overflow
- Shellcode
- No Operations (NOPs)
- Buffer Overflow Methodology
- Overflow using Format String
- Smashing the Stack
- Once the Stack is Smashed...
- Buffer Overflow Security Tools
- BoF Security Tool: BufferShield
- BoF Security Tools

Cryptography

- Cryptography Concepts
- Cryptography
- Types of Cryptography
- Government Access to Keys (GAK)
- Encryption Algorithms
- Ciphers
- Advanced Encryption Standard (AES)
- Public Key Infrastructure(PKI)
- Public Key Infrastructure (PKI)Certification Authorities
- Email Encryption
- Digital Signature
- SSL (Secure Sockets Layer)
- Transport Layer Security (TLS)
- Disk Encryption Tools
- Cryptanalysis Tool: CrypTool
- Cryptanalysis Tools
- Online MD5 Decryption Tool

Penetration Testing

- Pen Testing Concepts
- Security Assessments
- Security Audit
- Vulnerability Assessment
- Limitations of Vulnerability Assessment
- Introduction to Penetration Testing
- Penetration Testing
- Why Penetration Testing?
- Testing Locations
- Types of Pen Testing
- Types of Penetration Testing
- External Penetration Testing
- Internal Security Assessment
- Black-box Penetration Testing
- Grey-box Penetration Testing
- White-box Penetration Testing

CYBER SECURITY

Introduction to Cybercrime

- Definition and History
- Cybercrime and Information Security
- Who are Cybercriminals and their classifications
- The Legal Perspectives
- An Indian Perspective
- Cybercrime and the Indian ITAct 2000
- A Global Perspective on Cybercrimes
- Survival for the Netizens
- Concluding Remarks and Way Forward to Further Chapters

Cyberoffenses

- How Criminals Plan the Attacks
- Social Engineering
- Cyberstalking
- Cybercafe and Cybercrimes
- Botnets
- Attack Vector
- Cloud Computing

Cybercrime and Wireless Devices

- Proliferation of Mobile and Wireless Devices
- Trends in Mobility
- Credit Card Frauds in Mobile and Wireless Computing Era
- Security Challenges Posed by Mobile Devices
- Registry Settings for Mobile Devices
- Authentication Service Security
- Attacks on Mobile/Cell Phones
- Security Implications for Organizations
- Organizational Measures for Handling Mobile
- Organizational Security Policies and Measures in Mobile Computing Era
- Laptops

Tools and Methods Used in Cybercrime

- Proxy Servers and Anonymizers
- Phishing
- Password Cracking
- Keyloggers and Spywares
- Virus and Worms
- Trojan Horses and Backdoors
- Steganography
- DoS and DDoS Attacks
- SQL Injection
- Buffer Overflow
- Attacks on Wireless Networks

Phishing and Identity Theft

- Phishing
- Identity Theft (ID Theft)

The Legal Perspectives

- Cybercrime and the Legal Landscape around the World
- Cyberlaws in India
- The Indian IT Act
- Challenges to Indian Law and Cybercrime Scenario in India
- Consequences of Not Addressing the Weakness in Information Technology Act

- Digital Signatures and the Indian IT Act
- Amendments to the Indian IT Act
- Cybercrime and Punishment

Computer Forensics

- Historical Background of Cyberforensics
- Digital Forensics Science
- The Need for Computer Forensics
- Cyberforensics and Digital Evidence
- Forensics Analysis of E-Mail
- Digital Forensics Life Cycle
- Chain of Custody Concept
- Network Forensics
- Approaching a Computer Forensics Investigation
- Setting up a Computer Forensics Laboratory
- Computer Forensics and Steganography
- Relevance of the OSI 7 Layer Model to Computer Forensics
- Forensics and Social Networking Sites
- Computer Forensics from Compliance Perspective
- Challenges in Computer Forensics
- Special Tools and Techniques
- Forensics Auditing
- Antiforensics

Forensics of Hand-Held Devices

- Understanding Cell Phone Working Characteristics
- Hand-Held Devices and Digital Forensics
- Toolkits for Hand-Held Device Forensics
- Forensics of iPods and Digital Music Devices
- An Illustration on Real Life Use of Forensics
- Techno-Legal Challenges with Evidence from Hand-Held Devices
- Organizational Guidelines on Cell Phone Forensics

Cybersecurity and Organizations

- Cost of Cybercrimes and IPR Issues
- Web Threats for Organizations
- Security and Privacy Implications from Cloud Computing
- Social Media Marketing
- Social Computing and the Associated Challenges for Organizations
- Protecting People's Privacy in the Organization
- Organizational Guidelines for Internet Usage
- Safe Computing Guidelines and Computer Usage Policy
- Incident Handling
- Forensics Best Practices for Organizations
- Media and Asset Protection
- mportance of Endpoint Security in Organizations

Cybercrime and Cyberterrorism

- Intellectual Property in the Cyberspace
- The Ethical Dimension of Cybercrimes
- The Psychology, Mindset and Skills of Hackers and Other Cybercriminals
- Sociology of Cybercriminals
- Information Warfare

Cybercrime Case Study

- Real-Life Examples

- Mini-Cases
- Illustrations of Financial Frauds in Cyber Domain
- Digital Signature-Related Crime Scenarios
- Digital Forensics Case Illustrations
- Online Scams

Institute of Ai

Reach us at:

info@drixcat.com| +91-9953356262